Warrant

Verified-fact estate settlement, built for Alix

Every fact carries the sentence that justifies it. Verification is deterministic — no model is ever asked whether a model told the truth.

Team	Mitansh and Egor
Built for	Alix — Agents of Administration, Track 3 (with Track 1 intact)
Repository	github.com/mit37/warrant
Date	28 July 2026
Status	463 tests. Anvil live. Extraction run against a real model.

What this is, in one page

Warrant is a decision engine for estate settlement in which no claim reaches a filed document without evidence behind it. A model may propose any fact it likes; it cannot get that fact into a decision without producing a verbatim quotation that a deterministic checker locates in a document we actually hold. Facts that fail are quarantined — visible to a reviewer, unreachable by the rules engine.

463	4	58	171	41/41
TESTS	STATE PACKS	CA COUNTIES	FIELD MAPPINGS	FACTS VERIFIED

The four invariants

- **No fact without a warrant.** A quotation the checker cannot locate is quarantined.
- **No model in the decision path.** Models extract. Rules decide. The evaluator is pure.
- **No model arithmetic.** Totals are computed in code under a cited statutory rule.
- **Three-valued logic.** A predicate over a fact we do not hold is *unknown*, never false. A blocked rule names what it needs. Gap detection is free rather than a feature.

> **The problem in the mentor's own words**
>
> “Dear LLM, when you make a claim, you have to give me the verbatim quote. And then I have a deterministic system actually go in, check the transcript, and say, I need that exact quote to be in there. And if it's not, then I can't trust this claim.”
>
> Soren, Rules as Data track

What it does that nothing else here does

- **Onboards a government form it has never seen** — reads the words printed around each field geometrically, has a model propose the mapping, and refuses any mapping whose claimed label is not on the page.
- **Finds assets nobody mentioned** — infers a life policy from a $14.32 monthly debit that appears in no document.
- **Knows when its own answers expire** — every citation carries a re-check horizon, and where a statute publishes its amendment cycle, the exact date.
- **Declines, with a reason** — of twenty form/estate pairs, six are correctly withheld and one is blocked rather than answered.

Functionality, in full

1. Extraction and verification

Capability	What it does	State
Live extraction	Documents to candidate facts, each with the sentence it rests on	Run: 9 docs, 41 facts, 41 verified, $0.08
Deterministic verifier	Normalised exact match, then n-gram coverage. No model involved	Inherited, unchanged, load-bearing
Numeric guard	A quote whose figures are not in the document cannot verify	Added after a fabricated fee passed
Quarantine	Failed facts are visible and structurally unreachable by rules	Enforced by type, not convention
Record warrants	Facts imported from a case system cite system, record and field path	Re-read and checked on admission

2. Jurisdiction

Capability	Detail
Rule packs	California, New York, Pennsylvania, Texas — each sourced to the state's own legislature
County pack	All 58 California counties, with fee variation (San Francisco is $450, not $435)
Honest gaps	Each pack carries an explicit list of what could NOT be sourced
No fallback	An estate in an uncovered state gets 'no rule pack' and a hold, never another state's answers
Freshness	Every citation has a re-check horizon; scheduled statutory changes beat observed stability
Compiler	New counties compile from their own published text, model-proposed and quote-verified

3. Forms — Track 3

Capability	Detail
Geometry extraction	Widget rectangles plus the words printed around each, read off the page
Discovered mapping	Model proposes; the same matcher that checks facts checks the cited label
Refusal	Two proposals cited labels not printed where they claimed. Both recorded, not deleted
Adjudications	Human corrections live in an override layer, re-applied and re-verified
Applicability	Which forms apply is a rule with a citation, not a hardcode
Local fill	Real filled PDFs, every field read back
Anvil	Live. 139/139 bindings joined on three forms; values confirmed on the returned page

4. Discovery and obligations

Capability	Detail
Recurring detection	157 transactions to 12 recurring charges, across descriptor drift
Asset inference	Three assets found in no document: a life policy, a property policy, a safe deposit box
Suppression	An insurer already in the ledger is deliberately not re-reported
Hypotheses, not facts	Nothing inferred can enter the ledger. Enforced by a test
Shut-down board	Recurring charges still billing after death, with the cancellation path per vendor

5. Dispatch

Capability	Detail
Hold detection	Classifies the line from audio; transcription starts only when a person speaks
Handover	Pages the specialist the moment a human answers, with the script already resolved
Bounded speech	The agent may state only verified facts within the prepared scope
Consent map	Recording law per state; an unknown stops the call rather than defaulting to permitted
Mid-call dispatch	Documents the institution asks for are prepared and released on one approval

6. Audit and change

Capability	Detail
Dependency-tracked replay	Supersede one fact; only decisions that read it are re-evaluated
Provable skips	Decisions untouched by a change are shown not to have moved
Approval gate	Keyed on reversibility and blast radius, not on a blanket confidence score
Distribution hold	The one irreversible act, gated conservatively and for stated reasons

What has actually run

The distinction this document holds throughout: **measured** means a command in the repository produced it; **modelled** means it was calculated from assumptions that are named. Nothing here is an estimate presented as an observation.

Result	Figure	Evidence
Live extraction	41 proposed, 41 verified, 0 quarantined, 0 malformed	measured — recorded and replayable
Extraction cost	$0.08 for nine documents	measured
Field mappings	171 verified across 4 forms, 3 refused	measured
Anvil chain	139/139 bindings, values read back off the page	measured, live API
Filled PDFs	14 documents, 270 fields, all read back	measured
Forms correctly withheld	6 of 20 pairs, each with a stated reason	measured
Model agreement	87 of 94 fields; 3 of 7 disagreements were real errors	measured, two models
Recurring bleed found	$11,963/yr still leaving the estate	measured
Assets discovered	3, in no document	measured

The result we are least comfortable with, stated anyway

Two mappings passed verification and were still wrong. One put the signer's name in “Title, if applicable”; the other answered “are the assets in the custody of the court?” from a field meaning “does a court case exist”. Both cited labels genuinely printed on the form.

Locating a quote proves the model did not invent its evidence. It does not prove the reasoning is right. We found these by rendering the filled PDF and reading it — not by adding another model — and they are recorded as adjudications rather than quietly fixed.

Financial model

> **A distinction the pitch must not blur**
>
> The widely-quoted 570–900 hours is what the **family** spends. Alix's P&L turns on what its **specialist** spends. Reducing the first is marketing; reducing the second is margin. This model addresses the second and says so.

Inputs

Input	Value	Status
Fully-loaded specialist hour	$70	Alix's own figure — salary, benefits and payroll tax. Corroborated: BLS OEWS wages against the ECEC benefit load give $58 low, $74 central, $102 high, so $70 sits inside that range and a little below its middle
Alix revenue per estate	$9,000 minimum	Alix published pricing — 1% of assets, $9,000 floor
Probate referee commission (CA)	0.1%	Cal. Prob. Code 8961(a); minimum $75, maximum $10,000 under 8963
Specialist hours to onboard one form today	4 h	ASSUMPTION. Nobody has been timed doing it — this is the single figure most worth replacing

Per-estate saving, at $70/hr

Lever	Basis	Hours	Value
Form filling — 270 fields auto-populated	measured	2 – 4	$140 – 280
Hold time — 34 min per call, ~6 institution calls	modelled	3.4	$238
Jurisdiction lookup instead of research	modelled	2 – 5	$140 – 350
Recurring-charge shutdown	measured	1 – 2	$70 – 140
Appraisal — cash self-appraised under 8901	measured	—	$460 on the LA sample
Total per estate		8 – 14	$560 – 980

Against $9,000 of revenue that is **6 to 11 points of gross margin**, on a business whose alternative is hiring. And it compounds: the same rule pack and the same field map serve every subsequent estate in that jurisdiction at no additional cost, so the second estate in a county is cheaper than the first and the hundredth is nearly free.

The larger number: form onboarding

One-time per form, then amortised across every estate that ever touches that institution. This is the volume problem Track 3 was written around, and it is where the money actually is.

Scenario	New forms	Specialist time today	Cost at $70	With Warrant
Pilot	20	80 h	$5,600	model cost + review

Year one	200	800 h	$56,000	model cost + review
At scale	1,000	4,000 h	$280,000	model cost + review

Four hours per form is an assumption, not a measurement. Ian is the right person to replace it, and one sentence from him makes this table defensible in a way our own benchmark never will be.

Three-year shape

	Year 1	Year 2	Year 3
Estates processed	250	1,200	4,000
Per-estate saving at $770 midpoint	$192,500	$924,000	$3,080,000
New forms onboarded that year	200	400	900
Onboarding saving at $70/hr	$56,000	$112,000	$252,000
Total	$248,500	$1,036,000	$3,332,000
Jurisdictions covered	4 states	15 states	40 states

Estate volumes are illustrative. Alix does not publish how many estates it settles and we will not invent the figure — replace the top row with the real one and every row below it follows arithmetically. The saving rate and the hourly cost are the parts we stand behind.

Honest assessment

What is genuinely strong

- **The verification gate is real and load-bearing.** It rejects. A fabricated policy is quarantined; a substituted fee no longer passes; three form mappings were refused.
- **Form onboarding scales the way Alix needs.** No human placed a field. 258 widgets across four forms, and the marginal cost of the next form is a model call and a review.
- **The jurisdiction model is the right shape.** Compiled, cited, versioned, diffable, zero tokens at decision time — and it declines rather than guessing outside its coverage.
- **Asset discovery is genuinely novel.** Inference from payment traces, not extraction. The industry currently tells families to do this by hand.

What is weak, and by how much

Weakness	Severity	Detail
Four states of fifty	High	Architecture scales; content does not yet. Research-bound, not design-bound
Three courts researched	High	Lazy acquisition means we only need the counties in the book, but that is still a queue
Nobody has used it	High	No specialist has touched it. No user testing at all
Verification has a ceiling	Medium	Catches fabricated evidence, not wrong reasoning. Two confirmed misses
Voice never placed a call	Medium	Designed, tested, bounded — but no Twilio credentials, so unproven in the field
One estate, nine documents	Medium	Extraction is proven narrowly. No OCR'd scans in the loop
DL 142 does not join Anvil	Low	A property of the tool, not our map: detection on finds 38 fields against the real 52; detection off returns none, as it does not read the existing AcroForm. Filled locally, all 26 boxes
Economics are modelled	Medium	Two of five saving levers are calculated rather than observed, and the $70 hour is Alix's own figure corroborated from outside rather than measured inside

Is it feasible to put into production?

Yes, in the order below, and the first phase requires nothing from Alix's engineering team at all. The riskiest work is last on purpose.

Phase	Scope	Effort	Integration risk
1. Shadow run	Point the importer at real estates; produce plans and filled forms	1–2 weeks	None — reads their export format
2. Form onboarding	They hand over PDFs, we return verified maps and Anvil Casts	2–4 weeks	Low — their Anvil account
3. Jurisdiction service	Rule packs behind an API their system queries	4–8 weeks	Medium — a new dependency
4. Dispatch	Hold detection, mid-call document release	8+ weeks	High — needs legal sign-off

> **Where we would start**
>
> **Phase 2 pays for itself and touches nothing.** It is the highest return and the lowest risk, it addresses the deficiency Alix themselves named, and it needs no changes on their side beyond handing over a folder of PDFs.

How it compares

	What they do well	What is missing	Our position
Anvil	Excellent PDF fill and e-signature	Fields are placed by a human, per form	Complementary — we generate and verify the mapping they consume
EstateExec, Trust & Will	Consumer-facing checklists and guidance	Not specialist tooling; no jurisdiction engine	Different user entirely
Generic LLM plus RAG	Fast to stand up, broad coverage	No warrant, no effective dates, retrieval by similarity where you need lookup by key	This is the approach we deliberately rejected
Alix today	Real expertise, real clients, real book	Volume and defensibility, by their own account	Infrastructure underneath, not a replacement

What we need

Item	Why	Blocking?
Twilio credentials	The voice agent is built and tested but has never placed a call	Yes, for dispatch
SMTP / fax credentials	Mid-call document release	Yes, for dispatch
Ian: hours to onboard one form	Replaces the one assumption the onboarding table rests on	No, but it is the highest-value sentence available
Priority counties	Lazy acquisition means we only build what the book touches	No
Alix estate volume	Turns the three-year table from illustrative into real	No

Two things we would say out loud

The 900-hour figure does not survive its source. It traces to the founder's own estate, was later restated as a population average, and is now cited back by journalists as an industry estimate. It appears on at least six Alix pages with no citation. The nearest real research says 570, from a survey with a stated sample and error margin. We would use Alix's published range verbatim — 600 to 900 — because a range cannot be attacked as a false point estimate, and its low end brackets the survey.

We would not claim the verifier makes the system correct. It makes the system *accountable*, which is a smaller claim and a more defensible one. Everything this product does is designed so that when it is wrong, somebody can see exactly where and why.

> *“Tomorrow, if your system told me that I have to go through formal probate, you need to track me every decision that was made. You need to tell me where you got your legal information.”*
>
> — Soren, Rules as Data track

That is the whole specification, and it is the one we built to.